

Econometrics I

Lecture 8b: Event Studies

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Fall 2026

Two things called “event study”

- ▶ **The panel event study** (economics, marketing, strategy): some units experience an event — a policy, a product launch, an acquisition — at a known date. Trace the outcome *before and after*, relative to units that did not.
- ▶ **The returns event study** (finance, accounting): a firm announces something. Measure the stock's *abnormal* return around the announcement, relative to what the market model predicts.
- ▶ Same skeleton in both: re-index time so the event is at zero, build a counterfactual path from the pre-period, and read off the gap. What differs is where the counterfactual comes from.
- ▶ Plan: the panel version (with a preview of Week 9's trouble), then the returns version.
- ▶ Attribution: the questions that organize the first half — who is the control, what is normalized, what does a flat pre-period buy you — follow Goldsmith-Pinkham's Applied Methods lecture 14. The examples and the single-timing focus are ours.

Panel event studies

Setup: event time

- ▶ Units i , calendar periods t . Treated units all experience the event at the same date E ; the rest never do. (Staggered dates: next week.)
- ▶ **Event time** $k = t - E$. The period just before the event is $k = -1$.
- ▶ Potential outcomes $Y_{it}(0), Y_{it}(1)$. The object we want is a whole *path*:

$$\tau_k = \mathbb{E}[Y_{it}(1) - Y_{it}(0) \mid t - E = k, D_i = 1], \quad k = 0, 1, 2, \dots$$

the dynamic average treatment effect on the treated, k periods after the event.

- ▶ Why a path rather than a number? Effects ramp up (adoption takes time), fade (novelty), or reverse (intertemporal substitution). One coefficient hides all of that, and averages it with weights you did not choose.

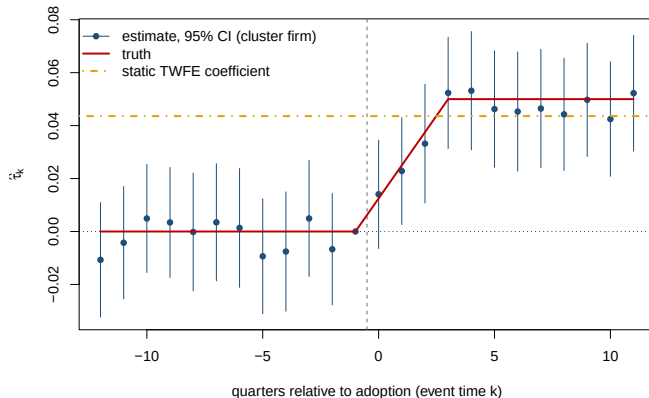
$$y_{it} = \alpha_i + \gamma_t + \sum_{k \neq -1} \tau_k \cdot \mathbb{1}\{t - E = k\} \cdot D_i + \varepsilon_{it}$$

- ▶ Two-way fixed effects (last hour) plus one dummy per event time, switched on only for treated units.
- ▶ **Reference period $k = -1$ is omitted.** Every τ_k is measured relative to the period before the event; $\hat{\tau}_{-1} \equiv 0$ by construction.
- ▶ Never-treated units have no event-time dummies. They pin down γ_t : what would have happened to everyone absent the event.
- ▶ Pre-period coefficients ($k < -1$) are the *diagnostic*: if the treated group was already diverging, they will show it. Cluster by unit.

Running example: an algorithmic-pricing tool

- ▶ 400 firms observed for 24 quarters. In quarter 13, half of them switch on a pricing-optimization tool. The outcome is log gross margin.
- ▶ Ground truth: the effect ramps from 0.0125 in the adoption quarter to 0.05 after three quarters and stays there. Nothing happens before adoption.
- ▶ Also in the data: firm effects α_i , a common macro path γ_t (random walk with drift), and noise. Treated firms are not a random half — but they are, in this first version, on the same trend as the others.
- ▶ Estimate the event-study regression with `fixest's i()`, cluster by firm, and plot.

The event-study plot



Flat and near zero before, ramping after, sitting on the truth. The dashed orange line is the *single* post-period coefficient from a two-way FE regression with one treatment dummy.

- ▶ The dot at $k = -1$ is zero *by construction*. Do not read it as evidence.
- ▶ The pre-period dots test “were the treated already moving relative to controls?” Here they are not, and the confidence intervals say how big a divergence we could have seen.
- ▶ The post-period dots are $\hat{\tau}_k$: the effect k quarters after adoption, relative to $k = -1$.
- ▶ The static two-way FE coefficient is a weighted average of the $\hat{\tau}_k$, $k \geq 0$, with weights proportional to how many observations sit at each k . With a ramp, it understates the long-run effect. With effects that fade, it overstates it. It is never the number you would have asked for.
- ▶ Confidence bands are pointwise. Twelve pre-period coefficients at 95% each is not a 95% test of “all flat.” Use a joint (Wald) test, or a uniform band.

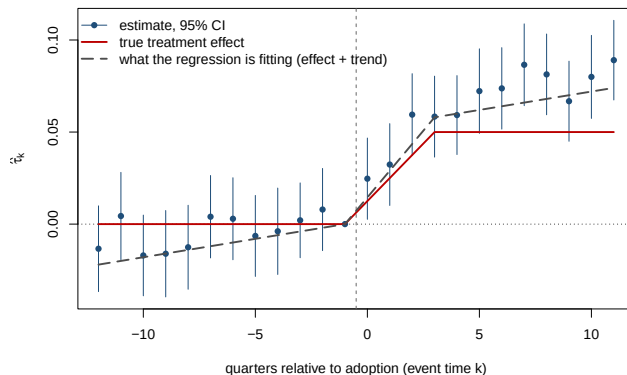
Assumptions, stated properly

1. **Parallel trends** in event time: absent the event, treated and control paths would have moved together,

$$\mathbb{E}[Y_{it}(0) - Y_{is}(0) \mid D_i = 1] = \mathbb{E}[Y_{it}(0) - Y_{is}(0) \mid D_i = 0] \quad \forall t, s.$$

2. **No anticipation**: $Y_{it}(1) = Y_{it}(0)$ for $t < E$. Units do not act on the event before it happens.
 - ▶ Under both, $\hat{\tau}_k$ is consistent for the dynamic ATT and $\tau_k = 0$ for $k < 0$.
 - ▶ Only the *implication* ($\hat{\tau}_k \approx 0$ for $k < -1$) is testable. Parallel trends in the post-period is an assumption about a counterfactual you never see.
 - ▶ Both are assumptions about *levels of $Y(0)$ differences*, not about levels of Y . Treated firms can be bigger, older, richer. They cannot be on a different trajectory.

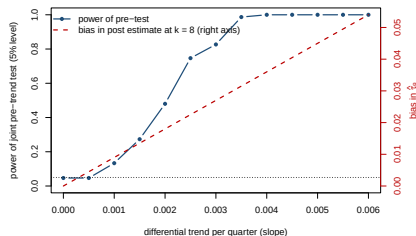
What a flat pre-period does not buy you



Same design, plus a differential trend of 0.2% per quarter for treated firms. No pre-period dot is individually significant and the joint test does not reject ($p = 0.30$). The post-period estimates are biased upward, and the bias grows with k .

The pre-test problem (Roth 2022)

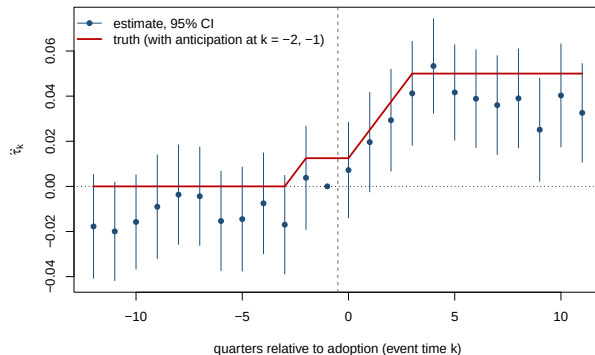
- ▶ A pre-trend test has power only against violations large relative to the noise. When the violation is small, the test passes and the bias is still there.
- ▶ Worse: **conditioning on passing** selects samples where pre-period noise offset the trend; the post-period estimates in those samples are biased *more*, not less.
- ▶ A linear trend with slope δ biases $\hat{\tau}_k$ by $\delta(k+1)$: the further out you look, the worse it gets, while the pre-test's power is fixed.



What to do instead

- ▶ **Plot, and report magnitudes.** “The pre-period coefficients are individually insignificant” is not a finding. “The data rule out a pre-trend larger than x per quarter, which would bias $\hat{\tau}_8$ by at most $9x$ ” is.
- ▶ **Sensitivity analysis.** Rambachan and Roth (2023): allow the post-period violation to be as large as (some multiple of) the largest pre-period violation, and report the range of τ_k consistent with that. HonestDiD does it in a few lines.
- ▶ **Longer pre-periods** help detect trends; they do not help if the trend is real.
- ▶ **Detrend only with care.** Fitting a treated-group trend on the pre-period and extrapolating it assumes the trend would have continued — a strong claim about a counterfactual, and it eats precision.
- ▶ **A better control group** is worth more than any of the above. Firms in the same segment, matched on pre-period trajectory, or — next week — a synthetic control.

Anticipation



Firms start adjusting two quarters before the tool goes live (pilot programs, contract renegotiation). The pre-period now shows a bump; and since $k = -1$ is the reference, every post-period estimate is shifted down by the anticipation effect. Fix: set the reference to $k = -3$, or redefine the event date as the announcement.

Design choices you will have to make

- ▶ **Who is the control?** Never-treated units are cleanest. If everyone is treated eventually, not-yet-treated units serve until their own event (staggered: next week).
- ▶ **No control group at all?** With a single event date and everyone treated, event time *is* calendar time. The event-time dummies are collinear with γ_t . `fixest` says “without doubt, your model is misspecified,” and it is right: you would be assuming the counterfactual is flat.
- ▶ **Window and binning.** Estimate τ_k for $k \in [-K_0, K_1]$ and bin everything outside into endpoint dummies, so far-out observations do not form a category of three firms.
- ▶ **Balance in event time.** If the panel is unbalanced, $\hat{\tau}_6$ comes from different firms than $\hat{\tau}_2$, and composition changes look like dynamics. Restrict to firms observed over the whole window, or say why not.
- ▶ **Reference period.** $k = -1$ by convention; earlier if anticipation is plausible.

Where it comes from: difference-in-differences in disguise

- Take two periods, $t \in \{E - 1, E + k\}$, and the two groups. The event-study coefficient is

$$\hat{\tau}_k = (\bar{y}_{E+k}^T - \bar{y}_{E-1}^T) - (\bar{y}_{E+k}^C - \bar{y}_{E-1}^C),$$

a 2×2 difference-in-differences comparing period k to the reference period.

- So the event study is a *collection* of DiD estimates, one per k , sharing a baseline. Everything you learn about DiD next week applies coefficient by coefficient.
- Equivalently: the regression is fitting each period's treated-minus-control gap and subtracting the gap at $k = -1$.

```
es[, k := ifelse(treated == 1, t - 13, -1)]           # controls get the reference value
fit <- feols(y ~ i(k, ref = -1) | firm + t, es, cluster = ~firm)
iplot(fit, ref.line = -0.5)                          # the event-study plot
wald(fit, keep = "k::-")                            # joint test on pre-period coefficients
```

```
es["k"] = np.where(es.treated == 1, es.t - 13, -1)
fit = pf.feols("y ~ i(k, ref=-1) | firm + t", data=es, vcov={"CRV1": "firm"})
fit.iplot()
```

- Binning: recode k to $\min(\max(k, -K_0), K_1)$ before $i()$.
- Never-treated units should carry the reference value, not NA: they must stay in the regression to estimate γ_t .

Preview of Week 9: staggered adoption breaks this

- ▶ Today every treated firm adopted in quarter 13. In practice adoption is **staggered**: cohorts $g = 2019, 2020, 2021, \dots$
- ▶ The same two-way FE regression then uses *already-treated* firms as controls for later cohorts. If effects are dynamic, those “controls” are moving, and the comparison is contaminated (Goodman-Bacon 2021; Sun and Abraham 2021).
- ▶ Symptoms: negative weights on some cohorts; pre-period coefficients that are non-zero even when there is no pre-trend; post-period coefficients that mix cohorts’ dynamics.
- ▶ Fixes (next week): estimate cohort-by-event-time effects and aggregate them yourself (Callaway–Sant’Anna, Sun–Abraham, Borusyak–Jaravel–Spiess imputation).
- ▶ The single-timing case is the one where two-way FE is exactly right. Know what “right” looks like before we break it.

Returns event studies

The returns event study

- ▶ Ball and Brown (1968), Fama, Fisher, Jensen and Roll (1969): does the stock market react to news, and how fast? Still the workhorse of empirical accounting and corporate finance.
- ▶ Typical events: earnings announcements, M&A announcements, CEO turnover, index inclusion, regulatory actions, product recalls, data breaches.
- ▶ The logic is the panel event study with one change: the counterfactual for firm i is not other firms, it is **firm i 's own relationship to the market**, estimated on pre-event data.
- ▶ The panel is long and narrow per event ($T \approx 250$ trading days per firm) rather than short and wide. So the “fixed effect” is estimated well, and the interesting inference is cross-sectional: averaging across events.
- ▶ Reference: MacKinlay (1997, *JEL*), “Event Studies in Economics and Finance.”

- ▶ Event day 0 for firm i . **Estimation window** $[-250, -11]$; **event window** $[-10, +10]$. The gap keeps the event out of the estimation sample.

- ▶ **Market model**, estimated on the estimation window by OLS, one firm at a time:

$$R_{it} = a_i + b_i R_{mt} + e_{it}, \quad \mathbb{E}[e_{it} \mid R_{mt}] = 0.$$

- ▶ **Abnormal return** on the event window: actual minus predicted,

$$\widehat{AR}_{it} = R_{it} - \hat{a}_i - \hat{b}_i R_{mt}.$$

- ▶ **Cumulative abnormal return** over a window $[k_1, k_2]$: $\widehat{CAR}_i(k_1, k_2) = \sum_{t=k_1}^{k_2} \widehat{AR}_{it}$.
- ▶ Compare to the panel version: a_i is α_i , $b_i R_{mt}$ is γ_t with a firm-specific loading, and AR_{it} is $\hat{\tau}_k$ for one firm.

Simulated example: earnings announcements

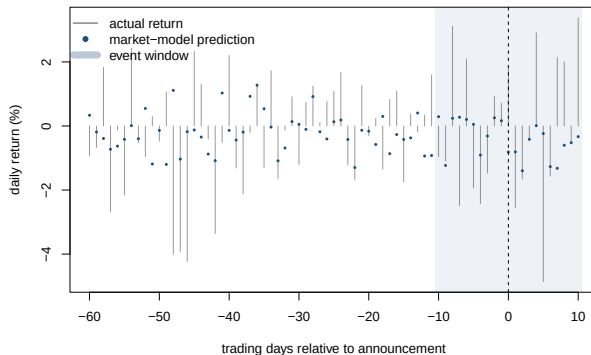
- ▶ 200 firms, each with one announcement. Daily returns for 261 days per firm.
- ▶ Market return $R_{mt} \sim N(0.04\%, 1\%^2)$, common to all firms on a given day. Betas $b_i \sim N(1, 0.3^2)$; idiosyncratic volatility 2% per day.
- ▶ Each firm has a standardized earnings surprise $S_i \sim N(0, 1)$. Ground truth for the abnormal return:

$$AR_{i0} = 0.02 S_i, \quad AR_{it} = 0.001 S_i \text{ for } t = 1, \dots, 10, \quad 0 \text{ otherwise.}$$

A one-SD surprise moves the stock 2% on the day and drifts another 1% over two weeks (post-earnings-announcement drift, Bernard and Thomas 1989).

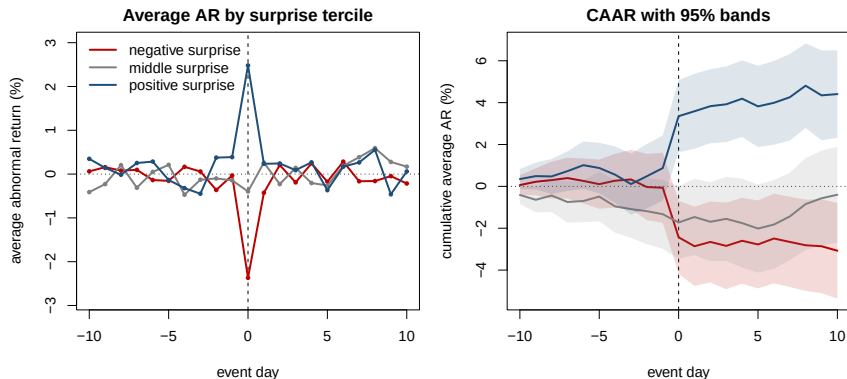
- ▶ So $CAR_i(0, 10) = 0.03 S_i$ in expectation, and $CAR_i(-10, -2) = 0$.

One firm



The firm with the largest positive surprise. Grey bars are actual daily returns; blue dots the market-model prediction. The abnormal return is the gap, and on day 0 it is hard to miss. On most other days it is noise of $\pm 2\%$: one firm tells you little.

Averaging across firms



Left: the average abnormal return by event day, by tercile of surprise. Right: cumulated. The day-0 jump and the post-announcement drift are both visible once you average over ≈ 67 firms. Nothing happens before day 0: the analog of a flat pre-period.

- ▶ Under the null of no abnormal return, $\widehat{AR}_{it}$ has mean zero and variance $\sigma_{e,i}^2 (1 + \text{small estimation-error term})$; over a window of length L , $\mathbb{V}[\widehat{CAR}_i] \approx L \sigma_{e,i}^2$.
- ▶ Two standard test statistics for the **average** CAR across N events:
 - **Cross-sectional t** : treat the N CARs as an i.i.d. sample. Robust to event-induced variance changes; needs N moderately large.
 - **Standardized** (Patell 1976; Boehmer et al. 1991): scale each firm's CAR by its own $\hat{\sigma}_{e,i} \sqrt{L}$ first, then average. Better power when volatility differs across firms.
- ▶ Both assume independence *across firms*. That fails when events cluster in calendar time (every firm announces in the same week; a regulation hits every bank on the same day). Then the market-model residuals are cross-sectionally correlated and the t -statistics are too large — the same clustering problem as Week 3, in a different costume. Fix: calendar-time portfolios, or cluster by event date.

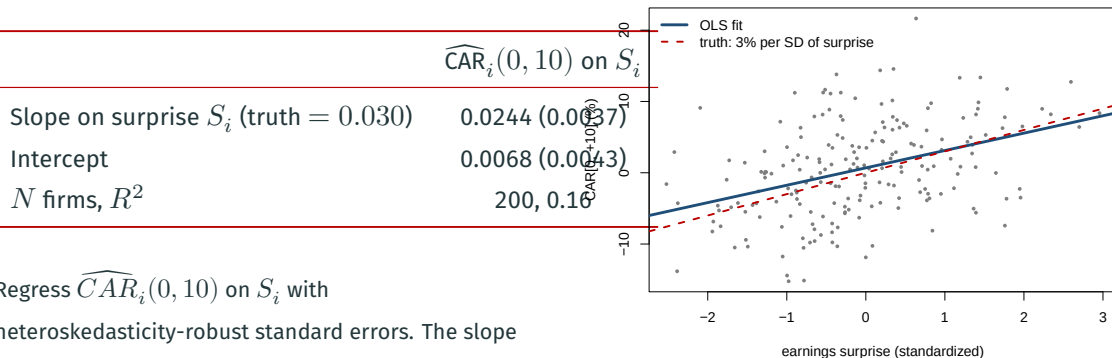
CARs by window

Window	CAAR (%)	t (cross-section)	t (standardized)	CAAR, $S > 0$	CAAR, $S < 0$
$[-10, -2]$	-0.25	-0.54	-0.57	0.17	-0.56
$[-1, 1]$	-0.02	-0.05	-0.08	2.71	-2.03
$[0, 0]$	-0.11	-0.53	-0.75	2.06	-1.71
$[1, 10]$	0.58	1.38	1.30	1.22	0.10
$[0, 10]$	0.47	1.01	1.02	3.28	-1.61

- The pre-event window is indistinguishable from zero; the announcement windows are not, once you split by the sign of the surprise. (Pooled over all firms, the average CAR is near zero because surprises average to zero. The *sign* of S_i is the treatment.)
- The two test statistics agree here because all firms have similar volatility.

The cross-sectional regression

The quantity accounting cares about is the **earnings response coefficient**: how much return per unit of surprise.



Regress $\widehat{CAR}_i(0, 10)$ on S_i with heteroskedasticity-robust standard errors. The slope recovers the truth. This is the CAR analog of putting a covariate in the event-study regression.

What can go wrong

- ▶ **Confounding events.** A dividend change announced with earnings; a merger rumor the day before. Short windows help; reading the news for each event helps more.
- ▶ **Event-date uncertainty.** Information leaks; after-hours announcements land on the next trading day. Widen the window, at the cost of noise.
- ▶ **The joint-hypothesis problem.** “Abnormal” is relative to a model. If the market model is wrong (missing factors, time-varying beta), so is the abnormal return. Over days it barely matters; over months or years (long-horizon “BHAR” studies) it dominates.
- ▶ **Thin trading and non-synchronous prices** bias $\hat{b}_i$ for small stocks.
- ▶ **Selection into the event.** Firms choose when to announce, whether to do a buyback, whether to be acquired. The CAR measures the market’s reaction *conditional on the firm having chosen the event*. That is often what you want; it is not the effect of assigning the event at random.

Recap

- ▶ An event study re-indexes time to the event and compares the outcome path to a counterfactual path.
- ▶ **Panel version**: two-way FE plus event-time dummies, reference $k = -1$, never-treated controls estimate γ_t . Needs parallel trends and no anticipation. The pre-period is a diagnostic with limited power; report what it rules out, not just whether it passed.
- ▶ **Returns version**: the market model is the counterfactual, estimated per firm on the pre-window. Abnormal returns, cumulated and averaged; cross-sectional t or standardized tests; watch for clustered event dates.
- ▶ Both are collections of difference-in-differences. Next week: what happens when the events happen at different times.

`inclass_session8.Rmd`, second half:

1. Reproduce the event-study plot and the joint pre-trend test. Then add anticipation and fix it by changing the reference period.
2. Reproduce the CAR table. Then move every firm's announcement to the same calendar day and add a common market shock on that day. What happens to the t -statistics, and why?